

CONVERTA SYSTEM DOCUMENTATION

Converta Comprehensive Documentation

Welcome to the official documentation for Converta, an ultra-modern, high-performance, in-brows

This document provides a complete overview of the website's architecture, design decisions, fea

Introduction & Philosophy

Converta was engineered to solve three major challenges in modern web-based file conversion:

- 1. Absolute Privacy: Standard conversion services upload user documents to remote servers, risk
- 2. Premium Aesthetic: Most utility websites are sterile, boring, and cluttered with ads. Conver
- 3. Hybrid Performance: Combining instant client-side processing for media with lightweight serv

Design & Neo-Precision Aesthetic

The user interface follows the Neo-Precision design systema tech-centric, hyper-clean theme eng

- Theme Palette:
 - Surface background: Deep Charcoal (#121508) for optimal contrast and reduced eye strain.
 - Primary Accents: Cyber-Lime (#b0d500) used for primary buttons, active progress indicators,
 - Glassmorphism: Navigation panels and bento action cards use multi-layered semi-transparent
- Typography: Uses Plus Jakarta Sans as the primary font, providing a sharp, modern geometric s
- Interactions: Dynamic hover states utilize subtle card expansions, shadows, and opacity trans
- Responsive Layout: Adaptive bento layouts shift fluidly between desktop grids and single-colu

Technical Architecture

Converta operates on a hybrid processing model:



- 1. Client-Side WASM Engine
 - Media files (audio, video) are converted entirely locally.
 - Technology: @ffmpeg/ffmpeg (WebAssembly wrapper).
 - Execution: Downloads the FFmpeg WebAssembly core directly into the browser. Files are loaded

- Benefit: Fast, secure, and utilizes client machine hardware without incurring server compute

2. Server-Side API Engine

Document operations and advanced image manipulations are processed via Next.js API routes.

- Technology: sharp for ultra-fast, multi-threaded image processing; pdf-lib for programmatic PDF
- Execution: Handled as serverless function routes (/api/convert).
- Benefit: Highly scalable, on-demand compute with zero-maintenance server structures.

File Structure

```
public/           Static assets (fonts, icons, favicon)
src/
  app/
    api/
      convert/
        route.ts  Node.js API endpoint for Serverless conversions
    dashboard/
      page.tsx    Interactive file conversion application & WASM engine
    globals.css   Tailwind v4 root CSS variables and Neo-Precision theme
    layout.tsx    Main wrapper (Plus Jakarta Sans font loading)
    page.tsx      High-fidelity marketing landing page
  components/
    ThemeToggle.tsx
    theme-provider.tsx
  package.json    Node.js dependency manager
  tsconfig.json   TypeScript configuration spec
```

Production & Deployment

- Hosting Platform: Vercel.
- CI/CD Pipeline: Connected to the GitHub repository (jagadeesvarrao-design/file-converter). An
- Performance Caching: Vercel automatically caches npm dependencies across builds, keeping comp
- Build Engine: Next.js compilation utilizes the Next.js compiler (Turbopack) for optimized bun